Repair and Refactor a Broken Python Application

Copilot in Practice

Session 27 | Intermediate | 3 hours

Agenda

Segment	Time
Repair as evidence gathering	8 min
Python failure order	10 min
Prepared repair demonstration	12 min
Recover behavior from fixtures	12 min
Lock behavior with tests	10 min
Refactor and lab handoff	8 min

The inherited application

The inventory tool should:

- read warehouse and ledger CSV files;
- normalize and aggregate SKU quantities;
- calculate stock variance;
- write stable text and JSON reports.

It currently does none of this reliably.

The broken-state contract

Stage	Named fault
Parsing	`SYN-01`, then `SYN-02`
Startup	`IMP-01`, then `IMP-02`
Behavior	`LOG-01` through `LOG-03`
Repeatability	`STATE-01`
Design	`DESIGN-01`

Each repaired fault reveals the next signal.

Start at the lowest blocked layer

```
parse
  ↓
import
  ↓
start
  ↓
behavior
  ↓
structure
```

Do not refactor code that Python cannot import.

Read the traceback from the bottom

Capture four facts:

1. exception type;
2. file and line;
3. failing source text;
4. command that produced it.

That is enough to form the first hypothesis.

Use Copilot inside a boundary

```
Explain this SyntaxError using only the shown file and traceback.
Identify the smallest syntax repair.
Do not refactor or change behavior.
```

The prompt limits the size of the proposed change.

Repair loop

```
capture failure
  → state one hypothesis
  → make one small change
  → run one narrow check
  → record the result
```

If the next failure changes, the repair probably moved execution forward.

Prepared demonstration

```
python -m inventory_reconciliation \
  --warehouse fixtures/input/warehouse.csv \
  --ledger fixtures/input/ledger.csv \
  --output-dir build
```

Expected first stop: `SYN-01` in `cli.py`.

Narrow checks beat noisy checks

Goal	Command
Parse package	`python -m compileall inventory_reconciliation`
Import entry point	`python -c "import inventory_reconciliation.cli"`
Check startup	`python -m inventory_reconciliation --help`
Check behavior	`pytest -q`

Use the smallest command that can disprove the current hypothesis.

Expected outputs are the contract

```
SKU     warehouse  ledger  variance  status  severity
AX-100         10      10         0  MATCH   none
BX-200          4       9        -5  SHORT   review
CX-300          0       8        -8  SHORT   urgent
```

Do not edit the contract to hide a defect.

Intended reconciliation rules

Rule	Expected behavior
SKU identity	Trim and uppercase
Duplicate rows	Sum after normalization
Missing side	Use quantity zero
Variance	Warehouse minus ledger
Output order	Sort by SKU
Repeated run	No prior results retained

Classification belongs in one place

Variance	Status	Severity
`0`	`MATCH`	`none`
`-1` to `-5` or `1` to `5`	`SHORT` or `OVER`	`review`
absolute value `6+`	`SHORT` or `OVER`	`urgent`

Duplicated rules drift. The starter proves it.

Deterministic defects give stable evidence

The same fixtures always expose:

- overwritten duplicate quantities;
- ledger-only SKUs that disappear;
- the wrong urgent threshold;
- stale results on a second call.

Stable evidence makes debugging discussions concrete.

Add tests before changing structure

Keep the fixture-level checks. Add focused tests for:

- severity at absolute variance `6`;
- negative quantity rejection with row context.

The refactor starts only after behavior is pinned down.

Global state creates delayed failures

```
RUN_HISTORY = []

def reconcile(...):
    RUN_HISTORY.extend(current_results)
    return RUN_HISTORY
```

The first call may look correct. The second call exposes the leak.

Tangled control flow hides policy

```
read rows
  + normalize values
  + merge quantities
  + classify variance
  + build output
  + retain state
```

Separate these jobs so each rule has one home.

Target design

```
cli.py
  → parsing.py
  → service.py
  → rules.py
  → reporting.py
```

The command-line layer coordinates. It does not own business rules.

Refactor in small moves

1. Remove process-wide mutable results.
2. Extract one classification function.
3. Reuse it from service and reporting.
4. Extract parsing and aggregation.
5. Run tests after each move.

Keep public behavior stable.

Review Copilot output like a patch

Ask:

- Did the change touch only the current fault?
- Which test proves the claimed behavior?
- Did a rule move or get copied?
- Does output order stay deterministic?
- What assumption is still unproved?

Generated code still needs evidence.

Lab workflow

Part	Time
Preflight and baseline	10 min
Restore parsing and imports	30 min
Recover behavior	30 min
Add regression tests	20 min
Refactor and verify	25 min
Final evidence	5 min

Final deliverable

- completed repair log;
- runnable inventory command;
- added regression tests;
- unchanged expected outputs;
- clean separation of parsing, rules, service, reporting, and CLI;
- passing `pytest -q`.

Lab handoff

Work in failure order. Record every repair.

Success looks like: the solution produces the supplied text and JSON outputs, passes all tests, and returns the same result on repeated runs.